

University Physics Quiz 6

Name: _____ SID: _____ School/Department: _____

1. The heat capacity of an ideal gas in a quasi-static (quasi-equilibrium) process is $C_x = \text{Const}$. Show that the

pressure p and volume V of the gas satisfy the equation $pV^f = \text{Const}$. in the process, where $f = \frac{C_x - C_p}{C_x - C_v}$.

(b) The energy density of the electromagnetic radiation (or the photon gas) is $u = \sigma T^4$ and its pressure is $p = u/3$, where T is the thermodynamic temperature and σ is a constant. Find the entropy density s of the electromagnetic radiation.

Solution: (a) From $dQ = C_x dT = dU + pdV = C_v dT + pdV = C_v dT + \frac{nRT}{V} dV$, we obtain

$$\frac{C_x - C_v}{nR} \frac{dT}{T} = \frac{dV}{V} \quad \text{or} \quad \frac{C_p - C_v}{C_x - C_v} \frac{dV}{V} - \frac{dT}{T} = 0.$$

Taking logarithm of both sides of the equation of state $pV = nRT$, we have $\ln p + \ln V = \ln nR + \ln T$. Differentiating

both sides we obtain $\frac{dp}{p} + \frac{dV}{V} = \frac{dT}{T}$. Substituting it into above equation, we have

$$\frac{C_p - C_v}{C_x - C_v} \frac{dV}{V} - \frac{dp}{p} - \frac{dV}{V} = 0 \quad \text{or} \quad \frac{dp}{p} + \frac{C_x - C_p}{C_x - C_v} \frac{dV}{V} = 0.$$

Integrating above equation we have $\ln p + \gamma \ln V = \text{Const}$. or $pV^f = \text{Const}$. (The process is called polytropic process.)

(b) $U = \sigma T^4 V$

$$\begin{aligned} dS &= \frac{dU + pdV}{T} = \frac{d(\sigma T^4 V) + \frac{1}{3} \sigma T^4 dV}{T} = \frac{\sigma T^4 dV + 4\sigma T^3 V dT + \frac{1}{3} \sigma T^4 dV}{T} \\ &= \frac{4}{3} \sigma T^3 dV + 4\sigma T^2 V dT = d\left(\frac{4}{3} \sigma T^3 V\right) \end{aligned}$$

Thus $S = \frac{4}{3} \sigma T^3 V + C$ (Obviously $S = 0$ if $V = 0$.) and $s = \frac{4}{3} \sigma T^3$.

3. A monatomic gas obeys the van der Waals equation $p = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}$ and its heat capacity at constant volume C_v is

$3nR/2$ in the limit $V \rightarrow \infty$.

(a) Show that $(\partial C_v / \partial V)_T = 0$.

(b) Determine the entropy of the system $S(T, V)$ up to an additive constant.

(c) Determine the internal energy of the system $U(T, V)$ up to an additive constant.

Solution: (a) From the fundamental equation of thermodynamics $dU = TdS - pdV$, we have

$$C_v = \left(\frac{\partial U}{\partial T} \right)_v = T \left(\frac{\partial S}{\partial T} \right)_v.$$

Then, from $dF = -SdT - pdV$ we have the Maxwell relation $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$ and therefore

$$\left(\frac{\partial C_V}{\partial V}\right)_T = T \left[\frac{\partial}{\partial V} \left(\frac{\partial S}{\partial T} \right)_V \right]_T = T \left[\frac{\partial}{\partial T} \left(\frac{\partial S}{\partial V} \right)_T \right]_V = T \left(\frac{\partial^2 p}{\partial T^2} \right)_V = 0.$$

(b) From $\left(\frac{\partial C_V}{\partial V}\right)_T = 0$, we know that $C_V = f(T)$. Using the condition $\lim_{V \rightarrow \infty} C_V = \frac{3}{2}nR$, we obtain $C_V = \frac{3}{2}nR$. Thus

$$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV = \frac{C_V}{T} dT + \left(\frac{\partial p}{\partial T}\right)_V dV = \frac{3nR}{2T} dT + \frac{nR}{V-nb} dV,$$

and $S = \frac{3nR}{2} \ln T + nR \ln(V-nb) + C = nR \ln(V-nb) T^{\frac{3}{2}} + C$.

$$(c) dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV = C_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV,$$

From $dU = TdS - pdV$, we have

$$\left(\frac{\partial U}{\partial V}\right)_T = T \left(\frac{\partial S}{\partial V}\right)_T - p = T \left(\frac{\partial p}{\partial T}\right)_V - p = \frac{nRT}{V-nb} - \frac{nRT}{V-nb} + \frac{n^2 a}{V^2} = \frac{n^2 a}{V^2}.$$

$$\text{Thus } dU = \frac{3nR}{2} dT + \frac{n^2 a}{V^2} dV \quad \text{and} \quad U = \frac{3nR}{2} T - \frac{n^2 a}{V} + C.$$

3. In the adiabatic throttling process (Joule-Thomson process), a definite amount of gas on the left side at pressure p_1 and of volume V_1 is slowly forced through a porous plug into right side at pressure p_2 and of volume V_2 .

(a) Show that the enthalpy is conserved in this process, i.e., $H_1 = U_1 + p_1 V_1 = H_2 = U_2 + p_2 V_2$.

(b) Show that the change in temperature for the gas in this process is given by

$$\Delta T = \int_{p_1}^{p_2} \frac{T^2}{C_p} \left(\frac{\partial(V/T)}{\partial T} \right)_p dp.$$

(Hint: Consider the Joule-Thomson coefficient $\mu_{JT} = \left(\frac{\partial T}{\partial p} \right)_H$.)

Solution:(a) The total work done by the pistons is

$$W = - \int_{V_1}^0 p_1 dV - \int_0^{V_2} p_2 dV = p_1 V_1 - p_2 V_2.$$

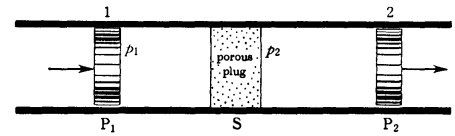
From the first law

$$U(p_2, V_2) - U(p_1, V_1) = W = p_1 V_1 - p_2 V_2.$$

Thus, we have

$$U(p_1, V_1) + p_1 V_1 = U(p_2, V_2) + p_2 V_2 \quad \text{or} \quad H(p_1, V_1) = H(p_2, V_2).$$

(b) Using the cyclic chain rule, we have



$$\left(\frac{\partial T}{\partial p}\right)_H = -\frac{\left(\frac{\partial H}{\partial p}\right)_T}{\left(\frac{\partial H}{\partial T}\right)_p} = -\frac{1}{C_p}\left(\frac{\partial H}{\partial p}\right)_T.$$

From $dH = TdS + Vdp = T\left(\frac{\partial S}{\partial T}\right)_p + \left[T\left(\frac{\partial S}{\partial p}\right)_T + V\right]dp$, we have

$$\left(\frac{\partial H}{\partial p}\right)_T = T\left(\frac{\partial S}{\partial p}\right)_T + V = V - T\left(\frac{\partial V}{\partial T}\right)_p = -T^2\left(\frac{\partial(V/T)}{\partial T}\right)_p. \quad (dG = -SdT + Vdp)$$

Thus, from

$$dT = \left(\frac{\partial T}{\partial p}\right)_H dp + \left(\frac{\partial T}{\partial H}\right)_p dH = \left(\frac{\partial T}{\partial p}\right)_H dp = \frac{T^2}{C_p}\left(\frac{\partial(V/T)}{\partial T}\right)_p dp.$$

($dH = 0$ The throttling process is the equal enthalpy process, as we have shown in (a).)

Therefore,

$$\Delta T = \int_{p_1}^{p_2} \frac{T^2}{C_p}\left(\frac{\partial(V/T)}{\partial T}\right)_p dp.$$

4. The differential of the Gibbs free energy can be written as $dG = -SdT + Vdp$.

(a) Show that $dQ = -T\left(\frac{\partial^2 G}{\partial T^2} dT + \frac{\partial^2 G}{\partial p \partial T} dp\right)$, and then show that dQ is not an exact differential.

(b) Show also that $C_p = -T \frac{\partial^2 G}{\partial T^2}$, $C_v = -T \left[\frac{\partial^2 G}{\partial T^2} - \left(\frac{\partial^2 G}{\partial p \partial T} \right)^2 / \frac{\partial^2 G}{\partial p^2} \right]$.

(c) Furthermore, show that $C_p - C_v = TV\alpha_v^2/\kappa_T$.

(Hint: The differential form $u(x, y)dx + v(x, y)dy$ is an exact differential if and only if $\partial u/\partial y = \partial v/\partial x$.)

Solution: (a) Since $dG = -SdT + Vdp$, we have

$$S(T, p) = -\frac{\partial G}{\partial T}, \quad V(T, p) = \frac{\partial G}{\partial p}. \quad (1)$$

Thus, we have

$$dS = \frac{\partial S}{\partial T} dT + \frac{\partial S}{\partial p} dp = -\frac{\partial^2 G}{\partial T^2} dT - \frac{\partial^2 G}{\partial p \partial T} dp,$$

and

$$dQ = TdS = -T\left(\frac{\partial^2 G}{\partial T^2} dT + \frac{\partial^2 G}{\partial p \partial T} dp\right).$$

Since $\frac{\partial}{\partial p} \left[-T \frac{\partial^2 G}{\partial T^2} \right] = -T \frac{\partial^3 G}{\partial p \partial T^2}$, $\frac{\partial}{\partial T} \left[-T \frac{\partial^2 G}{\partial p \partial T} \right] = -\frac{\partial^2 G}{\partial p \partial T} - T \frac{\partial^3 G}{\partial p \partial T^2}$, then $\frac{\partial}{\partial p} \left[-T \frac{\partial^2 G}{\partial T^2} \right] \neq \frac{\partial}{\partial T} \left[-T \frac{\partial^2 G}{\partial p \partial T} \right]$.

Therefore, dQ is not an exact differential.

(b) Consider a process at constant pressure $dp = 0$,

$$dQ = -T \frac{\partial^2 G}{\partial T^2} dT, \quad C_p = \left(\frac{dQ}{dT} \right)_p = -T \frac{\partial^2 G}{\partial T^2}. \quad (2)$$

By considering $p = p(T, V)$, we have $dp = \left(\frac{\partial p}{\partial T}\right)_V dT + \left(\frac{\partial p}{\partial V}\right)_T dV$.

Using the cyclic rule

$$\left(\frac{\partial p}{\partial T}\right)_V \left(\frac{\partial T}{\partial V}\right)_p \left(\frac{\partial V}{\partial p}\right)_T = -1,$$

we have

$$\left(\frac{\partial p}{\partial T}\right)_V = -\left(\frac{\partial V}{\partial T}\right)_T \bigg/ \left(\frac{\partial V}{\partial p}\right)_p = -\frac{\partial^2 G}{\partial p \partial T} \bigg/ \frac{\partial^2 G}{\partial p^2}.$$

(We also have $\left(\frac{\partial p}{\partial V}\right)_T = 1 \bigg/ \left(\frac{\partial V}{\partial p}\right)_T = 1 \bigg/ \frac{\partial^2 G}{\partial p^2}$.) Then, we have

$$\begin{aligned} dQ &= -T \left(\frac{\partial^2 G}{\partial T^2} dT + \frac{\partial^2 G}{\partial p \partial T} dp \right) \\ &= -T \left[\frac{\partial^2 G}{\partial T^2} - \left(\frac{\partial^2 G}{\partial p \partial T} \right)^2 \bigg/ \frac{\partial^2 G}{\partial p^2} \right] dT - T \frac{\partial^2 G}{\partial p \partial T} \bigg/ \frac{\partial^2 G}{\partial p^2} dV. \end{aligned}$$

In case of the constant volume, we have

$$-dQ = -T \left[\frac{\partial^2 G}{\partial T^2} - \left(\frac{\partial^2 G}{\partial p \partial T} \right)^2 \bigg/ \frac{\partial^2 G}{\partial p^2} \right] dT, \quad C_V = -T \left[\frac{\partial^2 G}{\partial T^2} - \left(\frac{\partial^2 G}{\partial p \partial T} \right)^2 \bigg/ \frac{\partial^2 G}{\partial p^2} \right]. \quad (3)$$

(c) From (2) and (3), we finally obtain

$$\begin{aligned} C_p - C_V &= -T \left(\frac{\partial^2 G}{\partial p \partial T} \right)^2 \bigg/ \frac{\partial^2 G}{\partial p^2} \\ &= -T \left(\frac{\partial V}{\partial T} \right)_p \bigg/ \left(\frac{\partial V}{\partial p} \right)_T = \frac{TV\alpha_V^2}{\kappa_T}. \end{aligned}$$